MIL-C-60028C (PA)
Amendment 1
30 January 1976

MILITARY SPECIFICATION

CARTRIDGE, 81MM, SMOKE, WP, M375 AND M375A2 ASSEMBLING, MARKING AND PACKING

This Amendment forms a part of Military Specification MIL-C-60028C (PA) dated 4 April 1969.

Wherever "M375" appears, add "and M375A2".

Wherever "8885264" appears, add "and 9240953".

Delete "M71A1E1" Percussion Primer wherever it appears and substitute "M71A2".

Paragraph 2.1, Delete "ET-888 5264 - Equipment Tabulation".

Paragraph 2.1, add the following:

"INSPECTION EQUIPMENT DRAWINGS

P9233991 - Inspection Device 81MM Mortar (Vert)
P9233992 - Inspection Device 81MM Mortar (Horiz) Opt.
P9242815 - Block Diagram Electrical
7257965 - Wear Limit Check Gage
6518071 - Profile and Alignment Gage
6518072 - Alignment Check Gage
9221731 - Strength Obturating Ring Heat Weld Test/Procedure

Delete paragraph "3.6 Strength of heat weld" in its entirety.

Paragraph 4.2.2.6 Cartridge, Critical defect "2. Ignition cartridge missing or inverted"; delete "Visual" and substitute "Test, dwg. P9233991, P9233992 and dwg. P9242815".

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Paragraph 4.2.2.6 Cartridge, delete Critical defect 4. "Fin broken or distorted" and substitute "Fins bent or damaged, or cartridge housing cracked".

Paragraph 4.2.2.6 Delete "Critical defect 5. Warning label missing or unidentifiable.....Visual". Add "Special defect, 100 Percent Inspection a. Warning label missing or unidentifiable.....Visual".

Paragraph 4.2.2.6 Cartridge, Major defects 108 and 109, delete "fails to meet disassembly torque" and substitute, for both defects, "fails to meet assembly torque without moving".

Paragraph 4.2.2.6 Cartridge, delete Major defect "114. Protective bag.....Visual 06036" in its entirety.

Paragraph 4.2.2.6 Cartridge, Major defect 116, Delete "on red end" and substitute "on red end or green end (as applicable)".

Paragraph 4.2.2.7 Add Major defect "104. Increment shells bind on inside diameter of container.....Visual-Manual 07004".

Paragraph 4.2.3.2, Delete "Equipment shall be in accordance with 4.2.4" and substitute "Inspection equipment shall be in accordance with dwg. 6518071, dwg. 6518072 and dwg. 7257965".

Paragraph 4.2.3.3 Proving ground test Delete "(d) Horizontal range less than 80 percent of mean horizontal range but not less than 70 percent of mean horizontal range (see 3.3.4.1), Special, Code No. 06018" in its entirety.

Paragraph 4.2.3.3 Proving ground test Delete "(e) Horizontal range less than 70 percent of mean horizontal range" and substitute "(e) Horizontal range less than 80 percent of mean horizontal range (calculated without the short rounds)".

Paragraphs 4.2.3.3.1 and 4.2.3.3.2 Delete the following: "The disposition of any lot, having cartridges with special defects, shall be made in accordance with paragraph 6.6".

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Paragraph 4.2.3.3.2 Delete "a sample of fifty cartridges from each lot" and substitute "a sample of forty-eight (48) samples from each lot".

Paragraph 4.2.3.5, Delete "3.6" and substitute "dwg. 8885264 or 9240953". In addition, delete "using test equipment in accordance with paragraph 4.2.4" and substitute "using test equipment in accordance with drawing 9221731".

Paragraphs 4.2.4, 4.2.4.1, 4.2.4.2 and 4.2.4.3, Delete these paragraphs in their entireties and substitute the following:

"4.2.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2."

Delete paragraph 4.3.1 "Proving ground" and substitute the following:

"4.3.2 Proving ground test.-The test shall be performed at a Government Proving Ground in accordance with Proving Ground Acceptance Test Procedure Supplement CR 81M-2J.

The cartridges shall be fired from a standard 81MM mortar at an elevation of forty-five degrees. One half of the sample cartridges shall be fired at charge one and the other half at charge nine. All fuzes shall be set for superquick functioning. The individual velocities, pressures, deflections and ranges shall be recorded on the firing record. Observation shall be made for misfires and duds, and shall be recorded. In the event of a misfire, the primer and ignition cartridge shall be removed and forwarded to Picatinny Arsenal. Misfired primers and ignition cartridges shall not be disassembled by the proving ground. Should a premature burst or metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The velocity and range standard deviations and arithmetic averages shall be calculated and recorded. Range probable error shall be calculated and recorded. Temperature of the day, weather and the time of firing shall also be recorded.

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Hawk radar shall be used to observe for metal parts separation, unobserved short rounds and unobserved duds. In these events, an analysis should be made of the radar data and should be noted on the firing record. This shall include a comparison of time of flight of the unobserved rounds with that of the normal rounds. Prints of the radar traces of those rounds, and prints of the preceeding and succeeding rounds, shall be attached to the firing record. Data from ballistically acceptable lots may be discarded. Smear camera shall be used to photograph all samples. The camera shall be positioned to pick up each round just outside the second velocity coil. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all the other rounds in that particular charge zone, shall be attached to the firing record. Film from acceptable lots may be discarded".

Paragraphs 6.2, 6.3, 6.3.1, 6.3.2, 6.3.3, 6.3.4 and 6.3.5, Delete these paragraphs in their entireties and substitute the following:

"6.2 Inspection equipment designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and Contractor Designs. SIE designs are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the Contractor. Equipment to be designed by the Contractor is designated "CD". The Contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the Contractor may submit alternate or modified Contractor designs of SIE in accordance with 6.2.2 and 6.2.3, should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the Contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the Contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs, if not otherwise covered thereby (See 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The Contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each Contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion data for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt."

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Delete paragraphs "6.6 Special defects, 6.6.1 Definition and 6.6.2 Disposition" in their entireties and substitute "6.6 Special Defects.-See MIL-A-48078 (PA) - Military Specification - Ammunition, Standard Quality Assurance Provisions, General Specification For."

Para. 4.2.2.7, delete Major: "103. Desiccant bag missing..... .....Visual 07003" in its entirety.

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